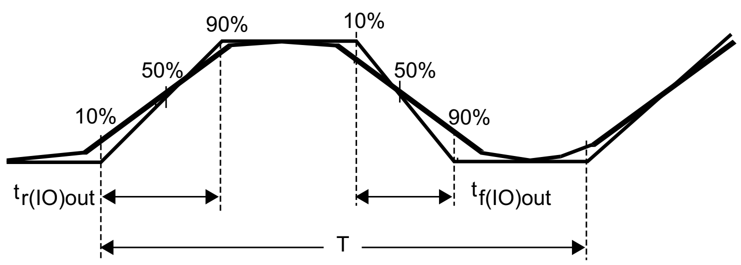
Figure 18. I/O AC characteristics definition(1)



Maximum frequency is achieved with a duty cycle at (45 - 55%) when loaded by the specified capacitance.

1. Refer to *Table 51: I/O AC characteristics*.

5.3.14 NRST input characteristics

The NRST input driver uses CMOS technology. It is connected to a permanent pull-up resistor, R_{PU}.

Unless otherwise specified, the parameters given in the following table are derived from tests performed under the ambient temperature and supply voltage conditions summarized in *Table 23: General operating conditions*.

Table 52. NRST pin characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{IL(NRST)}$	NRST input low level voltage	-	-	-	$0.3 \times V_{DD}$	V
$V_{IH(NRST)}$	NRST input high level voltage	-	$0.7 \times V_{DD}$	-	-	V
$V_{hys(NRST)}$	NRST Schmitt trigger voltage hysteresis	-	-	200	-	mV
R_{PU}(1)	Weak pull-up equivalent resistor(2)	$V_{IN} = V_{SS}$	25	40	55	kΩ
$V_{F(NRST)}$(1)	NRST input filtered pulse	$2.0\ V < V_{DD} < 3.6\ V$	-	-	70	ns
$V_{NF(NRST)}$(1)	NRST input not filtered pulse	$2.0\ V < V_{DD} < 3.6\ V$	350	-	-	ns

1. Specified by design. Not tested in production..
2. The pull-up is designed with a true resistance in series with a switchable PMOS. This PMOS contribution to the series resistance is minimal (~10% order).